Understanding Time-Based Data

A Foundational Tutorial Series for Public Safety Professionals

Beginner Level — Part I & II

20 Topics · Self-Paced · No Math Background Required

DRAFT VERSION

For internal review and discussion only

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Your feedback is highly valued. If you find any errors or have suggestions, please contact:

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How to Use This Tutorial Series

This series is designed for **public safety professionals, general audiences, and early-stage researchers** who want to understand time-based data without needing a statistics background. Each topic is a self-contained, one-page module — you can read them in order or jump directly to topics that interest you.

Who Is This For?

- **Public safety professionals** (law enforcement, analysts, administrators) who need to read and interpret crime or incident data
- **General public** readers interested in understanding how data tells a story over time
- **Early-stage researchers** building foundational intuition before moving to formal statistical methods

How It Is Organized

- **Part I: Foundations** (Topics 1–6) — What is time series data and how is it structured?
- **Part II: Patterns** (Topics 7–17) — What patterns can we see, and what do they mean?
- **Part III & Beyond** (Coming Soon) — Policy evaluation, causal inference, and forecasting

Reading Tips

- Each topic takes approximately 5 minutes to read.
- All data in examples are synthetic — created for teaching, not real incidents.
- Look for the 💡 Insight box and Key Takeaway at the end of each topic.
- Medium and Advanced levels, along with Python code exercises, are coming soon.

Questions or feedback? Contact: **yin.zhang@wsu.edu**

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Coming Soon: Part III, Part IV, and Python Code Exercises

PART I: FOUNDATIONS: UNDERSTANDING TIME-BASED DATA

Topics 1–6 | Estimated reading time: ~30 minutes total

This section introduces the core building blocks of time series data. By the end of Part I, you will be able to:

- Explain what a time series is and why it differs from a simple count or average
- Distinguish between time series data and cross-sectional (snapshot) data
- Understand how visualization, frequency, and rate calculations shape interpretation
- Recognize and handle missing data in a time-based sequence

All examples in this series use synthetic data from two fictional precincts. The data is designed to teach concepts, not to represent any real jurisdiction.

PART I: FOUNDATIONS

Topic 1: What Is a Time Series?

The Core Concept

A time series is a sequence of data points recorded at **successive, equally-spaced intervals**. In public safety contexts, it is not just about *how many* incidents occurred, but **when** they occurred in relation to each other.

Why It Matters

Imagine a Police Chief asks: "How are we doing with use-of-force?"

- **The single-number answer:** "We had 185 incidents this year." This tells you the volume, but not the story.
- **The time series answer:** "Incidents peaked in July and dropped by December." This reveals a narrative.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Precinct North, 2023 — Use-of-Force Incidents:

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

Looking at just the year-end average (~17 incidents/month) would hide the fact that incidents nearly doubled between January and July before declining — a pattern that demands investigation.

💡 **The Insight:** Time series data lets us see the narrative of public safety, not just the final score.

Key Takeaway: Always ask not just "how many" but "when" — the timing tells the real story.

PART I: FOUNDATIONS

Topic 2: Time Series vs. Cross-Sectional Data

The Core Concept

- **Cross-Sectional Data:** A "photograph" — a snapshot of many subjects at one point in time.
- **Time Series Data:** A "movie" — a recording of one subject across many points in time.

Why It Matters

If we only compare precincts at one moment in time (cross-sectional), we might conclude that Precinct South is "more dangerous" because its numbers are higher. But tracking each precinct over time (time series) reveals a very different story.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
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May	18	26	550
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- **Cross-Sectional View (December only):** North = 11, South = 26. Conclusion: "South is worse."
- **Time Series View (full year):** South stays between 22–28 all year — stable. North doubles from 12 to 25 — volatile.

💡 **The Insight:** Cross-sectional data helps answer "where to allocate resources today." Time series data answers "is a specific unit's situation improving or deteriorating?"

Key Takeaway: Use cross-sectional data for resource allocation; use time series data for policy evaluation.

PART I: FOUNDATIONS

Topic 3: Visualizing Time Series — The Line Plot

The Core Concept

The **line plot** is the gold standard for time series visualization. It connects data points with a line to emphasize **movement and direction**, making patterns instantly visible to the human eye.

Why It Matters

The human brain struggles to find patterns in a table of numbers, but it is expert at reading shapes. A line plot instantly reveals trends, seasonal waves, and outliers.

The Example

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For Precinct North, the table shows: 12, 10, 14, 15, 18, 22, 25... A line plot of these numbers would show a clear "mountain" shape — easy to interpret at a glance.

Common Visualization Mistakes

- **Using bar charts:** Bars emphasize individual values rather than the flow between them. They can obscure trends.
- **Hiding zero:** If the Y-axis does not start at zero, a small change can look like a massive crisis. Always check the Y-axis scale before drawing conclusions.
- **Plotting too many series:** More than 3–4 lines on one chart becomes difficult to read. Use separate panels if needed.

💡 **The Insight:** A well-designed line plot communicates more in 5 seconds than a data table communicates in 5 minutes.

Key Takeaway: Always visualize your data before analyzing it — the shape often tells you where to look.

PART I: FOUNDATIONS

Topic 4: Time Units, Frequency, and Aggregation

The Core Concept

- **Frequency:** How often we record data — hourly, daily, weekly, monthly, quarterly, or yearly.
- **Aggregation:** Grouping smaller time units into larger ones by summing or averaging (e.g., days → months).

Why It Matters

The frequency we choose can change our conclusions entirely:

- **Too granular (daily/hourly):** The data looks like random static. Every quiet night makes it look like crime has disappeared.
- **Too coarse (yearly):** A three-month spike that started and ended within the year becomes invisible.

The Example

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Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
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For Precinct North:

- **Yearly view:** "185 incidents total." The precinct seems consistently busy — no story.
- **Daily view:** "0, 1, 0, 0, 2, 0..." The data looks completely random and uninterpretable.
- **Monthly view:** The "mountain" pattern (winter low → summer peak → winter low) becomes clearly visible.

💡 **The Insight:** Monthly aggregation is often the "sweet spot" for public safety data — frequent enough to detect policy effects, smooth enough to filter out day-to-day noise.

Key Takeaway: Always match your time frequency to the question you are asking. Different frequencies can tell very different stories from the same data.

PART I: FOUNDATIONS

Topic 5: Counts vs. Rates Over Time

The Core Concept

- **Count:** The raw number of events (e.g., 25 use-of-force incidents in July).
- **Rate:** The count expressed relative to an activity baseline — e.g., UoF incidents ÷ total arrests.

Why It Matters

Use-of-force incidents in Precinct North peaked in July (25). Does this mean officers became more aggressive? Not necessarily — arrests also peaked at 700 in July. More police-public contact creates more *opportunities* for force, regardless of officer behavior.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	UoF Count	Arrests	Rate (UoF/Arrests)	Interpretation
Jan	12	400	3.0%	Baseline
Feb	10	380	2.6%	Slightly below baseline
Mar	14	450	3.1%	Roughly stable
Apr	15	460	3.3%	Stable
May	18	550	3.3%	Stable despite more arrests
Jun	22	600	3.7%	Slight uptick
Jul	25	700	3.6%	Peak count; rate nearly stable
Sep	19	500	3.8%	Rate slightly elevated
Oct	16	420	3.8%	Rate holding
Nov	14	390	3.6%	Returning toward baseline
Dec	11	350	3.1%	Near baseline

While the raw count more than doubled from January (12) to July (25), the rate stayed nearly constant — hovering around 3.0–3.8% throughout the year.

💡 **The Insight:** The count increase was mechanically driven by higher activity volume, not a change in officer behavior. Without the rate, this conclusion is invisible.

Key Takeaway: Always ask: "Is this change real, or is it just proportional to how busy we were?" Rates answer this question; raw counts cannot.

PART I: FOUNDATIONS

Topic 6: Missing Time Points and Reporting Gaps

The Core Concept

Time series analysis relies on a **continuous chain** of observations. A missing link does not just create a visual gap — it creates an unanswered question about what actually happened during that period.

Why It Matters

In public safety data, gaps commonly occur due to system upgrades, paper backlogs, or shifts in reporting standards. If we simply connect the dots across a gap, we are guessing — and our guess could be dangerously wrong.

The Example

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Precinct North is missing data for August. If we connect July (25) directly to September (19), we implicitly assume August was around 22. But consider the alternatives:

- **If a major protest occurred in August:** the real number could have been 100.
- **If the precinct was closed for renovation:** the real number could have been 0.

Before treating a gap as a smooth transition, always ask:

1. Why is this data missing? (System failure? Policy change? Staffing gap?)
2. Was the gap random, or does it coincide with a known event?
3. Can we recover the data from another source?

💡 **The Insight:** A "0" is a data point. A "Missing" is a question mark. Treat them very differently.

Key Takeaway: Never assume a gap is smooth. Investigate why the data is missing before trusting any trend that spans it.

PART II: PATTERNS EVERYONE CAN SEE (NO MATH REQUIRED)

Topics 7–20 | Estimated reading time: ~70 minutes total

Part I gave us the vocabulary. Part II teaches us how to read the story. This section covers the key patterns that appear in time series data — and, equally important, how to avoid common misinterpretations. By the end of Part II, you will be able to:

- Identify and distinguish between trend, seasonality, cycles, and noise
- Recognize outliers and understand how they distort averages
- Use smoothing and year-over-year comparisons to cut through noise
- Understand why policies take time to appear in the data
- Spot common data quality problems before they mislead your analysis
- Apply a systematic checklist when reading any time series chart

PART II: PATTERNS

Topic 7: Trend — The Long-Term Direction

The Core Concept

The **trend** is the overall, long-term direction of the data — ignoring short-term zig-zags. It tells you whether a situation is generally **improving, worsening, or stable** over months or years.

Why It Matters: One Bad Month ≠ Failure

A single high-incident month often triggers political or media attention. But if the long-term trend is downward, that spike may be a temporary fluctuation in an otherwise improving situation — not evidence of failure.

The Example

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Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
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	Nov–Dec	14, 11	24, 26	~370 avg

The summer spike in 2023 (June = 22, July = 25) looks alarming in isolation. But in context:

- 2021 summer peak: 42 incidents
- 2022 summer peak: 35 incidents
- 2023 summer peak: 25 incidents

The peaks are getting lower every year. The long-term trend is clearly downward, even though month-to-month, the pattern looks noisy.

💡 **The Insight:** The trend is the big picture. A single bad month does not erase years of improvement — and a single good month does not confirm that a crisis is over.

Key Takeaway: Always place the current month in the context of the last 2–3 years before drawing conclusions about direction.

PART II: PATTERNS

Topic 8: Seasonality — Predictable Patterns

The Core Concept

Seasonality refers to patterns that repeat at **fixed, predictable intervals** — every week, every quarter, or every year. Unlike noise, seasonal patterns are reliable and anticipatable.

Why It Matters

If we do not account for seasonality, we will misread routine fluctuations as crises — and waste resources reacting to predictable patterns we could have planned for.

The Example

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Look at Precinct North's July peak across three years:

- **2021:** 42 incidents (July peak)
- **2022:** 35 incidents (July peak)
- **2023:** 25 incidents (July peak)

The **timing** is identical every year — that is seasonality. The **level** is declining year over year — that is trend. Both patterns coexist.

Likely drivers: heat, longer daylight hours, more people outdoors, more police-public contact opportunities.

💡 **The Insight:** Understanding seasonality allows for proactive resource planning: schedule de-escalation training in May, increase staffing for summer events in June.

Key Takeaway: Do not overreact to predictable increases. Ask: "Is this higher than usual for this time of year?" — not just "Is this higher than last month?"

PART II: PATTERNS

Topic 9: Cycles vs. Seasonality

The Core Concept

- **Seasonality:** Fixed and predictable. Repeats every 12 months (or every week, quarter, etc.).
- **Cycles:** Not fixed. Can last 3–10 years. Driven by broad forces such as economic conditions, social movements, or multi-year reform efforts.

Why It Matters

Confusing a cycle with seasonality leads to bad predictions. If a city enters a 5-year economic downturn, crime may rise every year regardless of the season. If you treat this cycle like a seasonal pattern, you will be confused when numbers do not drop in winter as expected.

The Example

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Seasonality in this data: The summer peak (June–July) that repeats every year in Precinct North.

Cycle in this data: The overall decline from 2021 to 2023 — likely reflecting a multi-year reform effort or a broader societal shift — is a cycle. It operates on a scale of years, not months.

These two patterns overlap. The data can fall in summer (seasonal) while also declining year-over-year (cyclical). Analysts must mentally "layer" these patterns.

💡 **The Insight:** Seasonality answers: "What should we expect this month?" Cycles answer: "What direction are we heading over the next few years?"

Key Takeaway: Use seasonality for short-term scheduling and staffing. Use cycles for long-term budgeting, policy strategy, and resource planning.

PART II: PATTERNS

Topic 10: Noise and Irregular Fluctuations

The Core Concept

Noise is the "static" in data — random, unpredictable ups and downs that have no specific cause. It is what remains after you remove the trend, seasonality, and cycles. Every real-world dataset contains noise.

Why It Matters: Avoiding "Over-Explaining"

Not every data fluctuation requires a policy response. If use-of-force incidents go from 12 to 14 to 11, this might simply be random chance — not a crisis or a success. Trying to "explain" noise leads to chasing ghosts, wasting resources, and confusing staff.

The Example

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Look at Precinct South throughout 2021–2023: the data oscillates between 22 and 28 with no clear direction. There is no trend. There is no strong seasonal pattern. It is bouncing around a stable average of approximately 25.

A manager who investigates every month's fluctuation in Precinct South is chasing noise, not signal.

How do you tell noise from a real signal? Ask:

4. Is the change larger than the typical month-to-month variation we normally see?
5. Does it persist for more than 2–3 months?

6. Does it coincide with a known event, policy, or environmental factor?

💡 **The Insight:** A skilled analyst knows when to say: "This change is within the normal range of random variation." That statement is just as important as sounding an alarm.

Key Takeaway: Do not try to explain every small fluctuation. Focus on changes that are large, persistent, and plausibly linked to a real cause.

PART II: PATTERNS

Topic 11: Outliers and Spikes

The Core Concept

An **outlier** is a data point so far outside the normal range that it likely reflects a **specific, one-off event** rather than a structural shift in the underlying pattern. Unlike noise (which is small), a spike is massive and typically traceable to a single cause.

Why It Matters

Outliers can completely distort averages and mislead long-term analysis. Including them without flagging them gives a false picture of the underlying situation.

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In October 2023, Precinct South recorded 85 use-of-force incidents — more than three times the typical monthly average of 25. This was caused by a week of major civil unrest.

With the outlier included: Annual average for South = ~31. Looks like a worsening trend.

With the outlier identified and noted: Annual average for South = ~25 (excluding October). The precinct is actually stable.

Neither approach is dishonest — but they tell very different stories. The correct approach is to report both, with clear explanation.

💡 **The Insight:** When you see a spike, ask three questions: Is this the new normal (Trend)? Is this a regular occurrence (Seasonality)? Or is this a one-time event (Outlier)?

Key Takeaway: Always identify and label outliers before computing averages or making long-term decisions. Report them separately with context.

PART II: PATTERNS

Topic 12: Short-Term Fluctuations vs. Long-Term Change

The Core Concept

Short-term fluctuations are the month-to-month "zigs and zags" caused by weather, holidays, random chance, or local events. **Long-term change** is a structural shift in the data that persists across months and years.

Why It Matters

Reacting to short-term fluctuations as if they were long-term change is one of the most common errors in public safety data analysis. It leads to policy overreaction, wasted resources, and eroded trust in data-driven decision-making.

The Example

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Short-term view: Every summer, Precinct North's numbers go up. Every winter, they go down. A manager who only sees this year's data might think a summer crisis is a new problem.

Long-term view: The peaks are declining year over year (42 → 35 → 25) and so are the valleys (25 → 18 → 10). This is long-term structural improvement.

The same fluctuation looks like a "crisis" or "success" depending on whether you look at one month, one year, or three years. Context determines meaning.

💡 **The Insight:** Short-term fluctuations are the noise; long-term change is the signal. Always zoom out before you react.

Key Takeaway: Never evaluate a policy based on a single month's result. Always ask: "Where were we at this same time last year — and the year before?"

PART II: PATTERNS

Topic 13: Smoothing and Moving Averages

The Core Concept

Smoothing reduces the noise in data to reveal the underlying signal. The most common tool is the **moving average**, which replaces each data point with the average of itself and a set number of preceding points.

Why It Matters: Finding the Signal

When data is noisy, month-to-month changes can feel alarming even when nothing is really changing. A 3-month moving average helps answer: "Are we genuinely trending in a new direction, or is this just random variation?"

The Example

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Month	Raw Count (N. North)	3-Month Moving Avg	Interpretation
Jan	12	12.0	Baseline (only 1 month)
Feb	10	11.0	Slight dip — or just noise?
Mar	14	12.0	"Spike" smoothed away
Apr	15	13.0	Gradual upward movement
May	18	15.7	Trend becoming visible
Jun	22	18.3	Clear upward signal
Jul	25	21.7	Summer peak confirmed
Sep	19	—	Aug missing; avg skipped
Oct	16	17.3	Post-summer decline
Nov	14	15.7	Continuing downward
Dec	11	13.7	Returning to winter baseline

The raw count for March (14) looks like a 40% jump from February (10) — potentially alarming. But the 3-month moving average for March is 12.0, exactly where we started in January. The apparent spike was just the data returning to its mean after a quiet February.

Note: When August data is missing, the moving average for that month cannot be calculated — another reason missing data is problematic (see Topic 6).

💡 **The Insight:** The smoothed line tells you where the data is actually heading. The raw line tells you what happened on any given month.

Key Takeaway: Use a moving average when you want to answer "Are we better off now than we were three months ago?" — not "What happened last Tuesday?"

PART II: PATTERNS

Topic 14: Comparing Multiple Time Series

The Core Concept

Comparing multiple time series means placing two or more different groups on the **same graph over the same time period**. This allows us to determine whether changes are **local** (unique to one unit) or **systemic** (affecting everyone).

Why It Matters: Benchmarking and Context

If use-of-force incidents spike in one precinct, is it because of local leadership issues — or a city-wide factor like a heat wave, a major event, or a change in state law?

- **If both series move together:** The cause is likely external and shared (weather, city-wide policy, a protest).
- **If one series moves while the other stays flat:** The cause is likely internal and localized (a squad's tactics, a new commander, a neighborhood-specific problem).

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

In October 2023:

- Precinct North: 16 incidents (stable, near its seasonal average)
- Precinct South: 85 incidents (a massive spike)

Because we are comparing both precincts simultaneously, we immediately know that October's spike in South was not a city-wide event — it was localized. We do not need to investigate North's tactics; the problem is specific to South.

💡 **The Insight:** A comparison group is like a control group — it tells you what "normal" looks like during the same time window.

Key Takeaway: Always benchmark a specific unit's performance against a neighboring unit or city average. It separates local problems from systemic ones.

PART II: PATTERNS

Topic 15: Lagged Effects — Why Policies Don't Work Instantly

The Core Concept

A **lag** is a time delay between an action (such as a policy change or training program) and its visible effect in the data. In the real world, almost nothing takes effect immediately.

Why It Matters: Avoiding Premature Abandonment

A common mistake in public safety policy evaluation: a new de-escalation training program launches in January, numbers do not drop in February, and leadership declares the program a failure. But behavior change takes time — officers need to internalize new skills, supervisors need to reinforce them, and the effects need to accumulate across enough incidents to move the monthly average.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Hypothetical scenario based on Precinct North data:

Month	UoF Count	Training Status	Interpretation
Jan	22	✅ Launched	Training begins — no behavioral change yet
Feb	24	In progress	Short-term numbers may even rise (increased reporting awareness)
Mar	21	In progress	Still within normal noise range
Apr	18	Reinforced	Early signs of change — or just noise?
May	16	Reinforced	Downward signal becoming visible
Jun	14	Embedded	Clear effect: rate declining despite higher activity
Jul	17	Embedded	Summer spike — but lower than prior year (25 vs. 35 in 2022)

In this scenario, judging the program a failure in February would be premature — the actual effect only becomes visible from May onward, roughly 4–5 months after launch.

💡 **The Insight:** Expect a "slow-turning ship" effect from any new policy — not a reset button. The larger the organization and the more embedded the old behavior, the longer the lag.

Key Takeaway: When evaluating a new strategy, define the expected lag in advance (e.g., "we expect to see results by Month 4–6"). This prevents premature cancellation of effective programs.

PART II: PATTERNS

Topic 16: Autocorrelation — The "Memory" of Data

The Core Concept

Autocorrelation means that a data point is statistically related to its own past values. In plain terms, the data has **"memory"**: if incidents are high this month, they tend to be high next month too. The conditions that produced today's number have not disappeared overnight.

Why It Matters

High autocorrelation means the system has "momentum." Understanding this prevents two common mistakes:

- **Over-reacting:** Seeing a high month and assuming a new crisis when the data is simply carrying its prior level.
- **Over-optimism:** Seeing a low month and declaring victory when the system's momentum is still upward.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

Precinct South (Jan–May 2023): 25, 22, 28, 24, 26. The data stays "stuck" in a narrow band. February is influenced by the conditions that existed in January; March is influenced by February. The system is self-perpetuating.

Precinct North (Jan–Jul 2023): 12, 10, 14, 15, 18, 22, 25. This series also shows high autocorrelation — each month is close to the previous — but with an upward drift (trend).

Practical implication: if autocorrelation is high, the single best predictor of next month's number is this month's number. Large changes require significant interventions — the data resists sudden shifts.

💡 **The Insight:** High autocorrelation means "things tend to stay as they are." Changing this trajectory requires a sustained, significant force — not a one-time action.

Key Takeaway: If asked "what will our numbers be next month?" and the data shows high autocorrelation, the safest answer is: "Probably quite similar to this month."

PART II: PATTERNS

Topic 17: Stationarity — The Stable Baseline

The Core Concept

A time series is **stationary** if its key statistical properties — mainly its average (mean) and variability — remain constant over time. It has no trend and no growing or shrinking swings.

- **Stationary:** The data bounces within a consistent band around a fixed average. It looks the same wherever you look.
- **Non-stationary:** The average is moving (trend) or the swings are changing in size (growing volatility or seasonality).

Why It Matters: You Cannot Predict a Moving Target

Most advanced forecasting tools require stationary data. A non-stationary series must first be "stabilized" before predictions can be trusted.

Think of a basketball analogy:

- **Stationary player:** Scores around 20 points every game. Easy to predict: expect ~20 next game.
- **Non-stationary player:** Improving every week (trend) or scores 40 in winter and 5 in summer (seasonality). A simple average is useless as a prediction.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

- **Precinct South (excluding the Oct 2023 outlier):** Stationary. Stays between 22–28 throughout 2021–2023. Average always near 25. Predictable and stable.

- **Precinct North:** Non-stationary. Has a clear downward trend (declining average) and a recurring summer peak (seasonal variability). Both features make it non-stationary.

This distinction matters for forecasting: we can confidently predict Precinct South will be around 25 next month. For Precinct North, we must first account for the trend and the seasonal pattern before making any prediction.

💡 **The Insight:** Stationarity is a prerequisite for most formal forecasting. Before predicting, ask: "Is my data's average constant, or is it moving?"

Key Takeaway: If your data is stationary, the past is a reliable guide to the future. If it is not, you must account for trend and seasonality first.

PART II: PATTERNS

Topic 18: Year-over-Year (YoY) Comparison

The Core Concept

Year-over-Year (YoY) comparison means comparing a given period to the **same period in the prior year**. For example, comparing July 2023 to July 2022 — not to June 2023.

Why It Matters

This is one of the most practical tools in public safety data analysis. "How are we doing compared to last month?" is often the wrong question — it mixes up seasonal fluctuation with real change. "How are we doing compared to the same month last year?" **controls for seasonality** and reveals genuine progress or deterioration.

The question every commander, mayor, and police chief actually needs to answer: "Are we better off than we were a year ago at this same point?"

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	2021 (N. North)	2022 (N. North)	2023 (N. North)	YoY Change (2022→2023)	Signal
Jan	28	22	12	−45%	✅ Improving
Feb	26	20	10	−50%	✅ Improving
Mar	30	24	14	−42%	✅ Improving
Jun	40	32	22	−31%	✅ Improving
Jul	42	35	25	−29%	✅ Improving
Oct	30	24	16	−33%	✅ Improving
Dec	25	18	11	−39%	✅ Improving

Every single month in 2023 shows improvement compared to 2022 — and dramatic improvement compared to 2021. The YoY comparison makes this unambiguous, even during months that might look "high" compared to a January baseline.

Without YoY: Seeing July at 25 might cause concern. **With YoY:** July 25 (2023) vs. July 35 (2022) = a 29% reduction. That is a success, not a crisis.

💡 **The Insight:** Year-over-Year comparison is the simplest way to remove seasonal noise and show genuine directional change.

Key Takeaway: When presenting performance data to leadership, always include a YoY column. It frames the current month in the context that actually matters.

PART II: PATTERNS

Topic 19: Data Quality and Common Pitfalls

The Core Concept

Time series analysis is only as good as the data that feeds it. Public safety data has **unique quality challenges** that can create false signals — apparent trends or spikes that are artifacts of how data was collected, not reflections of what actually happened.

Why It Matters

A data quality problem that goes undetected can lead an entire organization to react to a crisis that does not exist — or to miss one that does. Recognizing these pitfalls is a fundamental analytical skill.

Common Pitfalls in Public Safety Data

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Pitfall	What It Looks Like	What To Do
System / Software Change	A sudden jump or drop with no real-world event	Check if a new RMS was deployed; compare to paper records
Definition Change	Category counts shift because the crime code was reclassified	Review policy memos; note the date of reclassification
Reporting Lag	Recent months look unusually low (records still being entered)	Flag last 1–2 months as provisional; revisit after 30 days
Under-Reporting	A low-crime period that coincides with low staffing	Cross-reference with arrest data or call-for-service logs
Outlier Contamination	One extraordinary event inflates the annual average	Identify, label, and report the outlier separately
Mixing Frequencies	Comparing monthly totals to a daily spike	Always match the time unit before comparing

A Diagnostic Checklist

When you see an unusual spike or drop, ask these questions before reacting:

7. Did we change record-management systems recently?
8. Did the definition or classification of this incident type change?
9. Are recent months' numbers still being entered (reporting lag)?
10. Is there a staffing gap that could explain lower reporting?
11. Is the outlier explainable by a single documented event?
12. Am I comparing compatible time units?

💡 **The Insight:** A "bad" month that coincides with a system upgrade is not a crime wave — it is a data artifact. Learn to tell the difference before escalating.

Key Takeaway: Before acting on any unusual pattern, verify that the data itself is trustworthy. Data quality is not a technical detail — it is a leadership responsibility.

PART II: PATTERNS

Topic 20: How to Read a Time Series Chart — A Checklist

The Core Concept

This final topic pulls together everything from Parts I and II into a practical, systematic checklist. When you pick up any time series chart — whether in a briefing, a report, or a news article — this is the order in which to read it.

The 12-Question Checklist

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

#	Question to Ask	What You Are Looking For
1	**What is the time unit?**	Hourly / daily / monthly / yearly? Match to your question.
2	**Is the Y-axis starting at zero?**	If not, visual changes may be exaggerated.
3	**Is there a trend?**	Long-term direction: up, down, or flat?
4	**Is there seasonality?**	Do the same peaks/troughs repeat each year?
5	**Are there outliers?**	Any point far outside the normal range?
6	**Is there missing data?**	Any gaps in the line? Investigate why.
7	**Are there multiple series?**	Do they move together (external cause) or apart (local cause)?
8	**What is the count vs. the rate?**	Does activity volume explain the change?
9	**What does last year look like?**	YoY comparison removes seasonal effects.
10	**Could there be a lag?**	Did a policy change precede this shift by 3–6 months?
11	**Is this trend or noise?**	Would a 3-month moving average change my conclusion?
12	**Could data quality explain this?**	System changes, definition shifts, reporting lags?

Putting It All Together

Good analysis does not require math. It requires asking the right questions in the right order. Every concept in this checklist corresponds to a topic in this tutorial series:

- Questions 1–2: Topic 3 (Visualization) and Topic 4 (Frequency)
- Questions 3–5: Topics 7, 8, 10, 11 (Trend, Seasonality, Noise, Outliers)
- Questions 6–7: Topics 6, 14 (Missing Data, Multiple Series)
- Questions 8–9: Topics 5, 18 (Rates, Year-over-Year)
- Questions 10–11: Topics 15, 13 (Lag, Moving Averages)
- Question 12: Topic 19 (Data Quality)

💡 **The Insight:** A structured reading habit turns raw charts into actionable intelligence. This checklist is your starting point — with practice, it becomes instinct.

Key Takeaway: Print this checklist. Keep it with your reporting toolkit. Use it every time you encounter a new time series chart.

Appendix: Glossary of Key Terms

All terms are defined in plain language for non-specialist readers. For deeper definitions, refer to the relevant topic in this tutorial series.

Term	Plain-Language Definition
Aggregation	Grouping smaller time units (e.g., days) into larger ones (e.g., months) by summing or averaging.
Autocorrelation	The degree to which a data point is statistically related to its own past values. High autocorrelation means the data has strong "memory."
Count	The raw number of events in a given period (e.g., 25 use-of-force incidents in July).
Cross-Sectional Data	A snapshot of many subjects at a single point in time (a "photograph").
Cycle	A pattern that repeats over a long, irregular period (3–10 years), such as economic downturns.
Frequency	How often data is recorded: hourly, daily, weekly, monthly, quarterly, or yearly.
Lag	A time delay between an action (e.g., a policy change) and its visible effect in the data.
Moving Average	A smoothing technique that replaces each data point with the average of itself and nearby past values.
Noise	Random, unpredictable fluctuations in data that have no specific cause. What remains after removing trend, seasonality, and cycles.
Non-Stationary	A time series whose average or variability changes over time (has a trend or seasonality).
Outlier	A data point so far outside the normal range that it likely reflects a one-off event rather than a structural shift.
Rate	A count expressed relative to an activity baseline (e.g., UoF incidents ÷ total arrests).
Reporting Bias	Systematic distortions in data caused by changes in recording systems, definitions, or staff behavior.
Seasonality	A pattern that repeats at fixed, predictable intervals (e.g., every 12 months).
Smoothing	Techniques (such as moving averages) used to reduce noise and reveal the underlying signal in data.
Stationarity	A property of a time series in which the mean and variance remain constant over time.
Time Series	A sequence of data points recorded at successive, equally-spaced intervals of time.
Trend	The long-term direction of a time series, ignoring short-term fluctuations.
Year-over-Year (YoY)	A comparison of a given period to the same period in the prior year, which controls for seasonal effects.

PART III: COMING SOON: POLICY EVALUATION & CAUSAL THINKING

How does time series meet decision-making? Part III will cover:

- Before-and-After Comparisons
- Interrupted Time Series Analysis
- Intervention Analysis — The "What If" Question
- Structural Breaks and Changepoints
- Correlation Over Time ≠ Causation

PART IV: COMING SOON: FORECASTING

How do we predict what comes next? Part IV will cover:

- Simple Forecasting Methods (Naive, Average, Trend-Based)
- Seasonal Decomposition
- Introduction to ARIMA Models
- Forecast Accuracy and Uncertainty
- Communicating Forecasts to Non-Technical Audiences

🐍 Python Code Exercises

Each topic in the Beginner, Medium, and Advanced series will be accompanied by a Python code exercise — no prior coding experience required. The exercises will be available as Jupyter notebooks on GitHub.

📹 CISER Video Tutorials

The Center for Interdisciplinary Statistical Education and Research (CISER) at Washington State University is also developing foundational statistics video tutorials led by subject-matter experts. These will be announced and linked on the GitHub repository as they become available.

Questions, feedback, or collaboration inquiries: **yin.zhang@wsu.edu**